

17 key controversy mapping notions

1. Post demographics (Rogers)

Is online friendship “real” friendship? Are social media accounts “real” people?

According to Richard Rogers, post-demographics is a way to study personal data in social networking platforms, focusing on how profiling is performed and its consequences, and departing from traditional demographic research methods.

The key takeaway for us is that the demographic way of doing sociology, which uses the person as a unit and focuses on race, ethnicity, age, income, or educational level, is not the only way to do sociology. We do not have to pretend that a user account matches a person, which is not true in general. We can still study user accounts sociologically.

In our case, we study the statements of various actors in the digital public space, without pretending that those statements are “representative”, in a traditional sense, of their thoughts. We can simply study the statements for what they are (traces).

Read more: Richard Rogers on post-demographics:

<https://www.digitalmethods.net/Digitalmethods/PostDemographics>

2. Just observe (Latour)

Bruno Latour, who shaped the pedagogical exercise of controversy mapping, famously stated that it boiled down to this:

"Just observe and describe controversies"

"just observe" refers to a methodological stance that emphasizes careful, detailed observation of controversies as they unfold, without imposing predetermined theoretical frameworks or rushing to resolve the disputes.

As argued by Tommaso Venturini, what makes it hard is precisely to refrain from the urge of writing your analysis using your favorite theories, your personal perspective on the subject, and your own certainties. Venturini's advice:

- Do not restrain your observation to any single theory or methodology;
- observe from as many viewpoints as possible;
- listen to actors' voices more than to your own presumptions.

Read more: Tommaso Venturini's "Diving in Magma" (2011), the main paper about this:

<https://www.tommasoventurini.it/wp/wp-content/uploads/2011/08/DivingInMagma.pdf>

3. Second degree objectivity (Latour)

Within a controversy, the absence of agreement between actors prevents us from finding “the objective truth.” But if there is no objectivity in controversies, how can they be studied objectively?

Second-degree objectivity acknowledges the impossibility of first-degree objectivity (complete neutrality) while still striving for a rigorous analysis. Second-degree objectivity means being objective about what actors say and do regarding disputed facts, rather than trying to objectively determine which facts are true.

Read more: Like above, Venturini's "Diving in Magma" (2011):

<https://www.tommasoventurini.it/wp/wp-content/uploads/2011/08/DivingInMagma.pdf>

4. Exploratory Data Analysis (EDA)

Exploratory vs. confirmatory: exploring data differs from applying the statistical toolbox to confirm hypotheses. Exploration finds questions, while confirmation finds answers.

Exploratory vs. explanatory: although the tools are the same, exploration entails visualizing for yourself while explanation entails visualizing for other people, which requires different strategies. Visualizing for yourself is about building internal knowledge and pivoting quickly between data facets. Visualizing for others requires communicating efficiently (more on this in lecture "visual storytelling").

Read more: This medium post showcases various techniques to explore data in Python with Pandas:

<https://medium.com/data-science/a-data-scientists-essential-guide-to-exploratory-data-analysis-25637eee0cf6>

5. Close / distant reading (Moretti)

Exploring data with algorithms and visualization can be seen as a form of reading. Franco Moretti distinguishes close reading and distant reading. Close reading is traditional reading, where the source material is not transformed. Distant reading entails transforming the source material to make patterns appear.

Your exploration of the Energy Islands project de-facto qualifies as a distant reading of what is said about the project in the digital public space.

Read more: A concise intro to distant reading by Brandon Walsh and Sarah Horowitz in their excellent book "Introduction to Text Analysis":

<https://walshbr.com/textanalysiscoursebook/book/reading-at-scale/distant-reading/>

6. Quali-quantitative analysis

Qualitative analysis typically focuses on interpretations and contexts. It typically uses interviews, observations (field notes), or long textual data. It looks into the richness and complexity of (social) phenomena, generally through close reading.

Quantitative analysis is focused on measurements and statistical patterns. It typically uses surveys, datasets, and experimental data. It aims to identify general trends and correlations, i.e. distant reading.

Quali-quantitative analysis combines both: it seeks to preserve the richness of qualitative insights while leveraging the scalability and structure of quantitative data. In digital sociology, this often means analyzing large data sets in a way that maintains attention to meaning, context, and narrative. In other words, it aims to provide thick description at scale.

Quali-quantitative analysis is not quali and quanti at the same time, but alternates qualitative moments and quantitative moments, close reading and distant reading.

Read more: Once again, Venturini's "Diving in Magma" (2011):

<https://www.tommasoventurini.it/wp/wp-content/uploads/2011/08/DivingInMagma.pdf>

7. Principle of symmetry (Bloor)

The principle of symmetry is an element of David Bloor's "strong programme" for the sociology of scientific knowledge. It demands that the same types of explanations are used for successful and unsuccessful knowledge claims alike.

Our methods should be able to analyze bad science as well as good science, whatever that means. We should be able to understand the knowledge claims of flat earthers as well as the knowledge claims of renowned physicists.

Of course, that does not mean that we consider them the same (see "provide weighing" below).

Read more: the Wikipedia page about the strong programme is insightful: https://en.wikipedia.org/wiki/Strong_programme

8. Actor (in actor-network theory)

In this actor-network theory (ANT), an actor is someone or something that makes a difference to the situation. Not every person is an actor: people who do not make a difference are not considered actors. We can also have non-human actors, like organizations (agencies, companies, associations...) but also animals, rivers, technologies... Non-humans can be considered actors insofar as they make a difference.

The key takeaway for this course is that we are mainly interested in actors insofar as we can track and document which difference(s) they make to the Energy Island project and the discourses about it. Here, an actor is an actor because it makes knowledge claims (e.g., involving expertise or risk) that influence decisions, answer open questions, shape the public debate, mobilize new actors, or make a difference in any other way.

Read more: Bruno Latour's "Where Are the Missing Masses?" (1992) explains it well:

<http://www.bruno-latour.fr/sites/default/files/50-MISSING-MASSSES-GB.pdf>

9. Follow the actors (Latour)

In accordance with the principle of symmetry, we do not investigate according to our own beliefs and knowledge, because this way we would not be able to account for the knowledge claims of actors we disagree with. Instead, we follow the actors by listening to their knowledge claims and how they argue about them to retrace their underlying logic. We potentially follow the actors to uncomfortable places we disagree with, and we account for it.

In practice, it means we pay special attention to the specific terms and arguments used by the different actors. We do not replace their words by ours, even when we think they are the same (because to them, they may not be the same). This is especially relevant when using queries to investigate a dataset.

Read more: for a more in-depth discussion about the symmetry principle and what it means to follow the actors in the digital public space, check Noortje Marres and David Moats' "Mapping Controversies with Social Media" (2015):

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2567929

10. Weighing positions

Some actors are stronger and make a bigger difference than others. You must show it.

The principle of symmetry states that we *treat all actors equally*, that we describe them fairly regardless of whether or not we agree with them. But this does not mean that we pretend that they are equal, in fact quite the contrary. Our analytical job is precisely to account for the fact that some of them make a bigger difference than others. For example, their knowledge claims were repeated by more other actors, or convinced more policy makers to take action, or were more influential on what topics are debated on social media, etc.

Following second-degree objectivity, provide an objective description (a measurement) of each actor's ability to make a difference to the Energy Islands project and the discussions about it.

11. Measurements

In sociology, measurements refer to the practical work of making social life knowable, comparable, and actionable within different research communities (or networks of practice).

The visualizations and numbers we obtain from the dataset through the notebooks constitute measurements.

From a post-demographic perspective, we recognize that our corpus does not *represent what people say* about Energy Islands. We neither measure nor aim to measure "what people say" about the project. However, our measurements can be compared, contrasted, and contextualized with one another to generate valid insights, even though the corpus lacks traditional representativeness.

12. Circulating reference (Latour)

Latour's "circulating reference" underscores that scientific facts don't just mirror reality directly, but instead gain their power to represent the world by moving through chains of transformations where each step changes the format but maintains the connection.

Here is a way to describe our analysis of the Energy Islands project as a chain of reference:

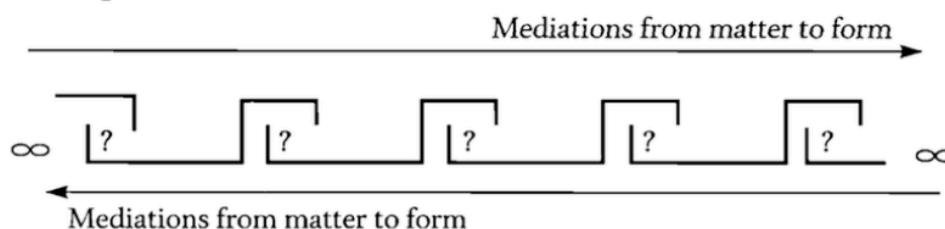
1. What people say over the internet is transformed into a series of web documents to harvest
2. These web documents are transformed into a structured table of actor statements (with metadata)
3. The statements in this table are transformed into homogeneous, normalized statements (translation...)
4. The normalized statements are transformed into data points we enrich with algorithms (semantic map, clusters, issues coding...)
5. The dataset is used as source material to produce measurements and visualizations using the notebooks
6. The visualizations and measurements are used as source material to ground arguments in a policy brief

Your policy brief will be well grounded insofar as your argument, through an unbroken chain of transformation, actually refers to what people say over the internet. A single broken link, such as misusing a visualization, can make your entire argument fall apart.

Read more: Bruno Latour's chapter "Circulating Reference" in Pandora's Hope (1999):

<http://www.bruno-latour.fr/sites/default/files/downloads/53-PANDORA-TO-POFIL-pdf.pdf>

Circulating reference



13. Truth and robustness

When engaged in the production of scientific knowledge, we must recognize that our arguments do not become robust because they are inherently true; rather, they start becoming true as they get more robust. In other words, we must invest considerable effort in consolidating our arguments. It is not enough for us to be convinced of the validity of our analysis; we must also demonstrate to others what makes our reasoning resilient to criticism and counterarguments.

This is especially important when analyzing a controversy, where competing versions of "truth" emerge from disagreements among different actors. In such contexts, we must not only be capable of deconstructing others' arguments to examine their foundations and assess their robustness, but we must also be able to construct compelling arguments of our own.

Here, the key takeaway is that solid arguments have a specific structure. This structure lets your audience easily follow your reasoning: they can see what evidence you're using and understand how you're building your case from that evidence.

Read more: Bruno Latour's chapter "Opening Pandora's Black Box" in Science in Action (1987):

<https://thehangedman.com/teaching-files/svd-phd/1-latour/latour-SIA-0-introduction.pdf>



14. The Toulmin approach to building an argument

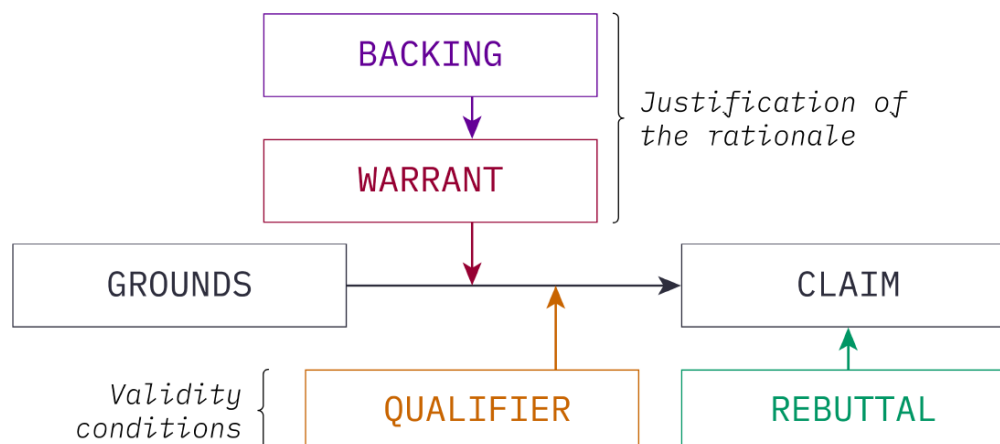
The Toulmin model was introduced by British philosopher Stephen Toulmin in 1958. It helps us understand how arguments are structured and categorized. There are six parts to it:

- **Claim.** A statement that something is so.
- **Grounds.** The backing for the claim: data and measurements (among other possibilities).
- **Warrant.** A bridge-like statement explaining why the claim follows from the grounds. General rule justifying the claim given the grounds.
- **Backing.** Support or explanation provided for the warrant. Many kinds are possible depending on the field.
- **Qualifier.** The degree of certainty offered for the claim. The modality.
- **Rebuttal.** Exceptions to the claim.

The key takeaway is that a robust argument is not just a claim backed by grounds (i.e., data). It is also, and perhaps more importantly, about the relation between the grounds and the claim, which Toulmin calls the "warrant" and the "backing". To make better arguments, learn how to make your warrant and backing explicit.

Read more: an accessible introduction to the Toulmin method:

https://owl.purdue.edu/owl/general_writing/academic_writing/historical_perspectives_on_argumentation/toulmin_argument.html



15. Technical mediation

In Science and Technology Studies (STS), mediation is a core concept that challenges the idea that technology simply transmits or carries information neutrally from one point to another. Technologies are better understood as mediators rather than mere intermediaries, because in order to analyze how they shape controversies, we need to observe that they translate, modify, and redirect the interactions they facilitate. **In short: technical mediation is non-neutral.**

There is another side to this. As we use tools (ex: algorithms) to process data and make measurements, mediations pile up. Each step is another link in the chain of the circulating reference, and a mediation whose non-neutrality we need to account for. When we produce a visualization, we need to be able to explain what is going on for each mediation involved.

Read more: Bruno Latour's "On technical mediation" (1994):

<http://www.bruno-latour.fr/sites/default/files/54-TECHNIQUES-GB.pdf>

16. Explanatory visualization for visual arguments

Explaining requires a different strategy than exploring, because you need to communicate context to your audience. Even when you use the same tools (for example, the notebooks).

Visual communication is an entire field that we cannot summarize in a few words. A classic advice would be to prioritize clarity about anything else, and after that to minimize distortions. Indeed, visualization is a mediation, so it is not neutral, and introduces information loss and distortion; nevertheless, those should be mitigated as much as possible.

When it comes to building a robust argument in a policy brief, you need to unpack the mediation for the reader. The same way the Toulmin model emphasizes the necessity to make the *warrant* explicit to connect the ground to the claim, you must highlight the visual pattern you ask the reader to see, and you must explicitly state why your reading follows from the visual pattern. This constitutes the context the reader needs to be communicated to follow your visual argument.

17. Annotation (of data visualizations)

After an exploration phase, your brain is loaded with context about the data. But when you transition to communicating your results to an audience, you need to externalize this internalized knowledge so that your audience has access to it. Your visualizations must feature a title, a legend, textual explanations, all of which can communicate the context. But your best tool is to *annotate* the visualization.

Annotation is a way to guide your reader's eyes to the visual pattern you need them to see. The textual content of the annotation allows you to state what the pattern means. Your text explanations should also state with clarity how your interpretation follows from the visual pattern (like a warrant does in the Toulmin system) which may require some backing (again, like in the Toulmin system).

As an annotation points to a key visual pattern, it is always better *in-place*, i.e. close to the pattern itself.

Read more: 10 good practical tips about annotating data visualizations:
<https://speakingppt.com/10-rules-for-graph-annotations/>

